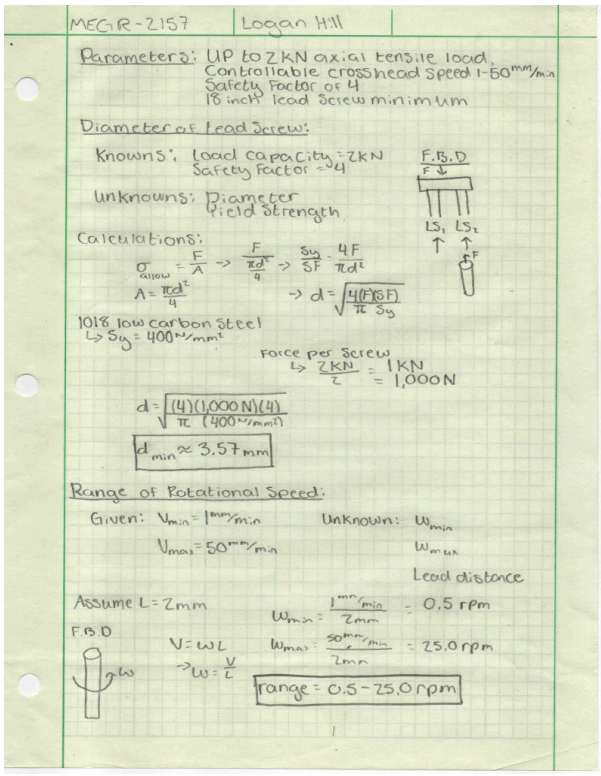
# Assignment 10: 2157 Students

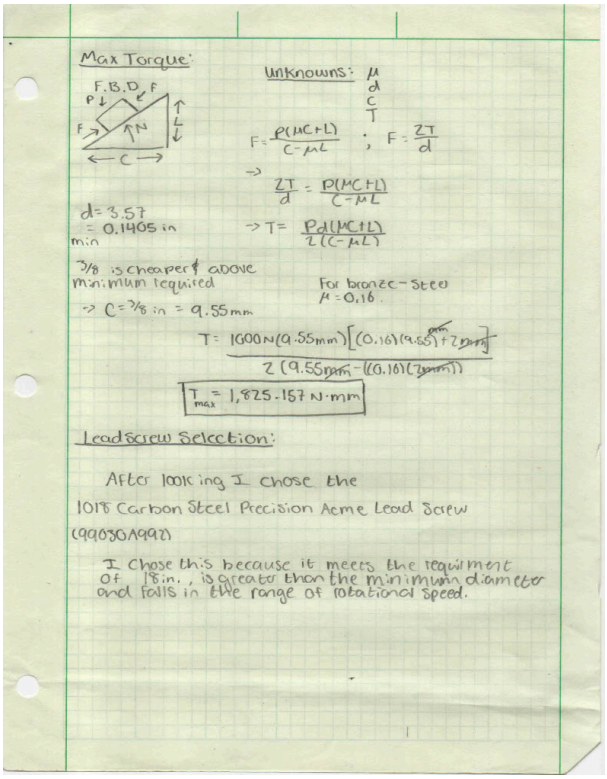
**Evan's Team:** Logan Hill, Evan Morrow, Levi Elliott

## Introduction

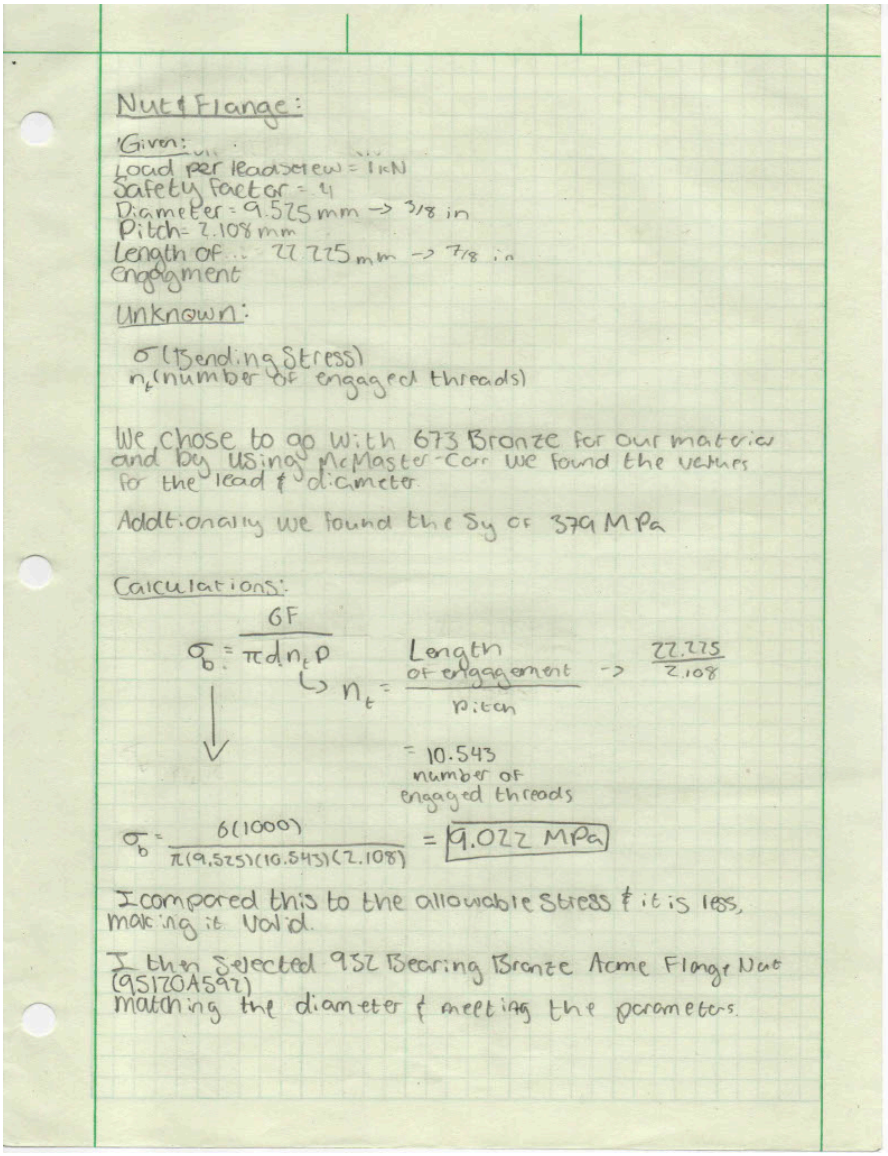
This week we are tasked to conceptual design a tensile tester using a lead-screw-drive crosshead. We will design and build a mechanically driven tensile tester capable of applying up to a 2kN axial tensile load with a controllable crosshead speed from 1 to 50 mm/min. We are told to use no electronics or software, purely mechanical applications for this part of the assignment. We are given a safety factor of 4 and a 18 inch leadscrew minimum parameter to work with.

## **Leadscrew Diameter** - Logan Hill



## **Nut and Flange** - Logan Hill



Nut & Flange:

Given:
Load per leadscrew = 1 kN
Safety factor = 4
Diameter = 9.525 mm → 3/8 in
Pitch = 2.108 mm
Length of Engagment = 27.275 mm → 7/8 in

Unknown:

σ (Bending Stress)
$n_t$ (number of engaged threads)

We chose to go with 673 Bronze for our material and by using McMaster-Carr we found the values for the lead & diameter.

Additionally we found the Sy of 379 MPa

Calculations:

$$\sigma_b = \frac{6F}{\pi d n_t p} \quad \hookrightarrow n_t = \frac{\text{Length of engagement}}{\text{Pitch}} \rightarrow \frac{27.275}{2.108}$$

$= 10.543$ number of engaged threads

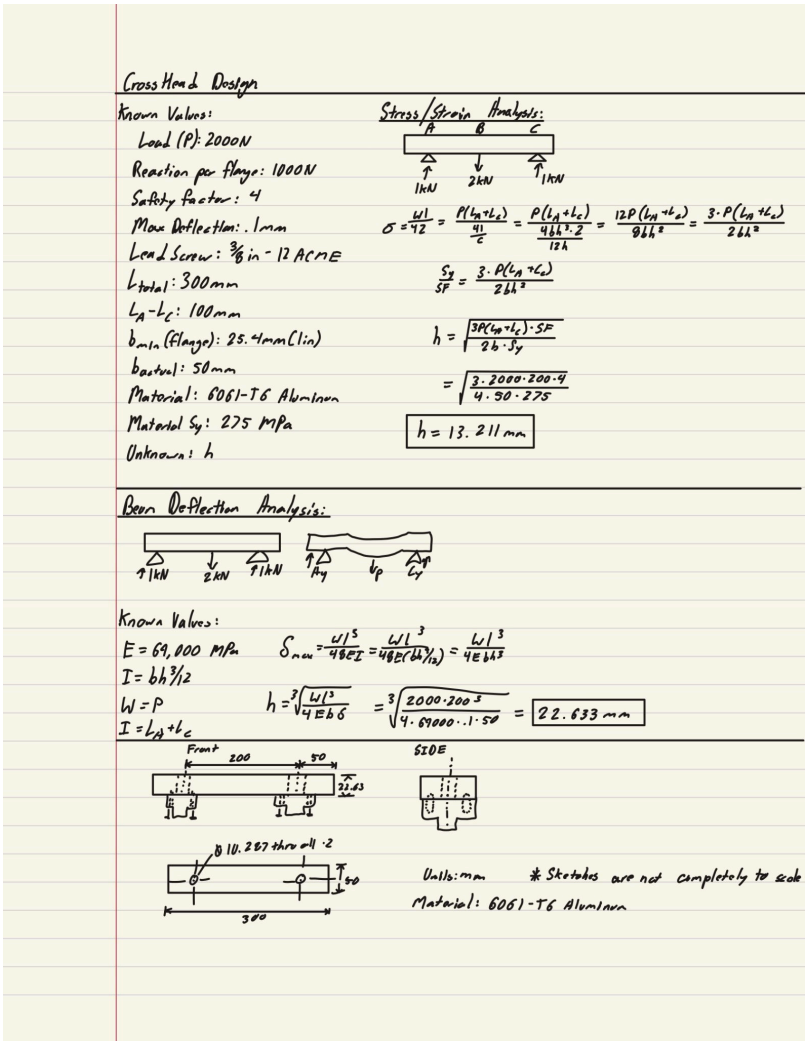
$$\sigma_b = \frac{6(1000)}{\pi(9.525)(10.543)(2.108)} = \boxed{9.022 \text{ MPa}}$$

I compared this to the allowable stress & it is less, making it valid.

I then selected 932 Bearing Bronze Acme Flange Nut (95170A592) matching the diameter & meeting the parameters.

## **Crosshead Design** - Evan Morrow

Using the known and unknown values I calculated for the necessary height of the crosshead to withstand the parameters of the 2000N. I did Stress/Strain and Beam Deflection Analysis on the crosshead and then made the engineering sketches as specified in the requirements. Work available below:



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## **Cost of Materials** - Logan Hill

| Item | Quantity | Part Number | Cost/Item | Total Cost | Link |
|---|---|---|---|---|---|
| Lead Screw | 2 | 99030A992 | $17.04 | $34.08 | https://www.mcmaster.com/99030A992/ |
| Flange Nut | 2 | 95120A592 | $111.78 | $223.56 | https://www.mcmaster.com/95120A592/ |
| T-Slot Frame | 1 (10ft) | 47065T107 | $86.54 | $86.54 | https://www.mcmaster.com/47065T107-47065T978/ |
| ½-20 Screw | 1 | 92210A744 | $3.09 | $3.09 | https://www.mcmaster.com/92210A744/ |
| Clevis Pin | 2 | 98306A915 | $1.82 | $3.64 | https://www.mcmaster.com/98306A915/ |
| ½-20 Flange Nut | 1 | 91094A116 | $1.38 | $1.38 | https://www.mcmaster.com/91094A116/ |
| M3 Bolt | 6 | 92095A201 | $0.19 | $1.14 | https://www.mcmaster.com/92095A201/ |
| M3 Hex Nut | 6 | 90591A250 | $0.03 | $0.18 | https://www.mcmaster.com/90591A250/ |

| | | | | | |
|---|---|---|---|---|---|
| Load Cell | 1 | 6131N47 | $270.68 | $270.68 | https://www.mcmaster.com/6131N47/ |
| Clevis Rod | 2 | 4749T151 | $10.17 | $20.34 | https://www.mcmaster.com/4749T15-4749T151/ |
| Gusset Bracket | 2 | 47065T799 | $15.10 | $30.20 | https://www.mcmaster.com/47065T799/ |
| Flange Bearing | 2 | 2827N5 | $4.20 | $8.40 | https://www.mcmaster.com/2827N5/ |
| **NET COST:** | | | | $683.23 | |

For cost analysis I looked at what parts were needed to construct our tensile tester and found the respected part in Mcmaster-Carr. From here I tabulated the cost along with the number of components needed to calculate the total cost of the assembly. I attached the links and part numbers for reference.

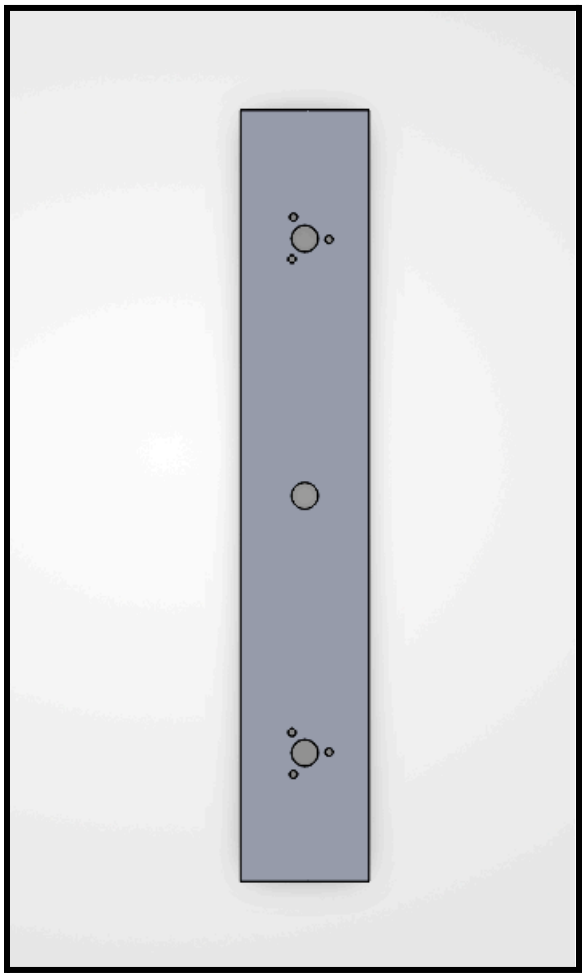
# Assignment 10: 2157 Students

## **Parametric Modeling** - Evan Morrow

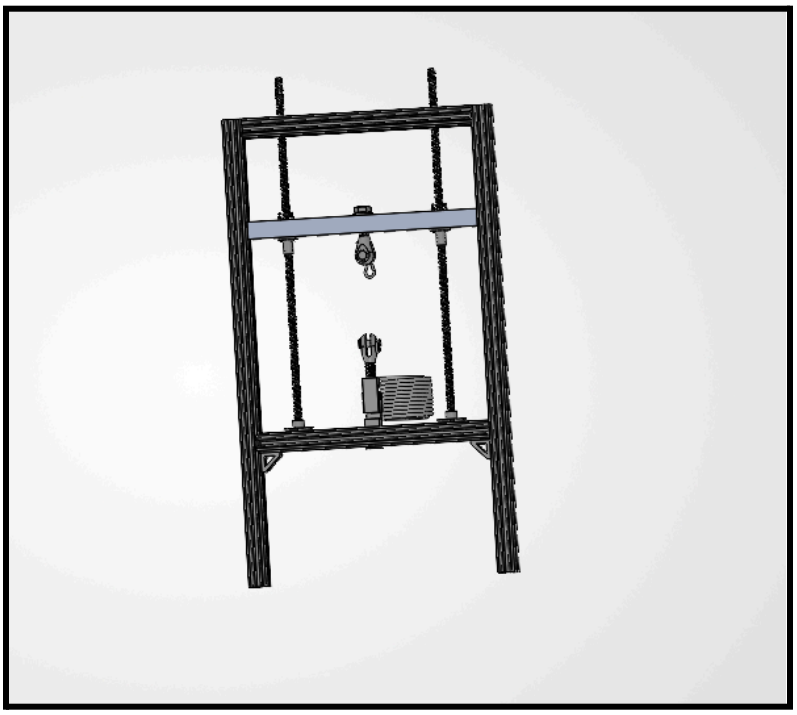
For the parametric modeling took the parameters from the calculations given above and inputted the equations and values into the equations in SolidWorks.

| Name | Value / Equation | Evaluates to |
|---|---|---|
| **Global Variables** | | |
| "Sy_head" | = 275 | 275.00 |
| "SF" | = 4 | 4.00 |
| "b_xhead" | = 50 | 50.00 |
| "P" | = 2000 | 2000.00 |
| "R" | = 1000 | 1000.00 |
| "dmax" | = .1 | 0.10 |
| "spacing_xhead" | = 200 | 200.00 |
| "h_xhead" | = 100 | 100.00 |
| "E_xhead" | = 69000 | 69000.00 |
| "I_xhead" | = ( "b_xhead" * ( "h_xhead" ) ^ 3 ) / 12 | 4166666.67 |
| "h_xh_stiffness" | = ( ( "P" * ( "spacing_xhead" ) ^ 3 ) / ( 4 | 22.63 |
| "h_xh_strength" | = ( ( "SF" * 3 * "P" * "spacing_xhead" ) / ( | 13.21 |
| "I_head" | = 300 | 300.00 |
| "screw_od" | = .375 | 0.38 |
| "fit_clearance" | = 10.287 | 10.29 |
| "m3_major_diam" | = 3 | 3.00 |
| "m3_hole_placement" | = .375 | 0.38 |

I then took the equations and created the crosshead. I made the three holes across the same diameter than added the screw holes around the 2 outer circles to secure the lead screws to.



I then inserted the parts from the cost analysis and pieced together the parts. This was the most challenging part as I was dealing with many dynamic surfaces like the t shaped brackets. I then finished the assembly and created a pack and go to submit it below.



A10 Assembly.zip

## Lessons Learned

### Logan Hill:

During this project I learned how to calculate the required lead screw diameter, the required speed and the required maximum torque to achieve the desired tensile load. I gained an understanding of how lead screw dimensions and thread geometry affect the force and speed of the system. I also learned how to analyze the thread bending stress of a bronze flange nut and to ensure the bronze flange nut can safely support the applied load with the required factor of safety. I learned how to research and select lead screws and flange nuts comparing lead screw materials, flange nut materials, specifications and costs from various suppliers. Overall this project improved my understanding of how engineering calculations, component selection and cost analysis work, in mechanical design. I spent about 4-5 hours on this assignment.

### Evan Morrow:

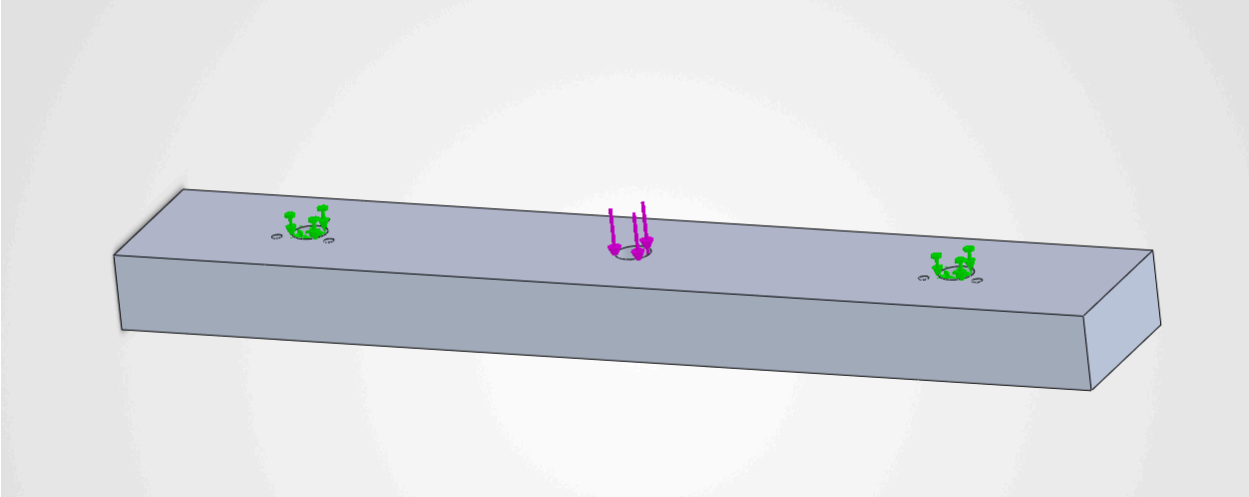
I learned the importance of collaboration and how it can be helpful in completing a larger project. I believe it is necessary for engineers to know how to work and communicate with others and what expectations are. The biggest challenge for me was assembling everything together as many of the parts do not go together easily and mating the parts takes some additional time. This project took me about 4 hours to complete from delegation to completion of my parts and portfolio.

### Levi Elliott:

Throughout this assignment I learned how important a team is when it comes to designing. With a team there is more brainstorming and it is important to consider everyone's considerations and each person's thought process. Communication with each other was very important with getting the work done in a timely manner. The toughest part of the assignment for me was creating the drawing. This is because I had to think about the necessary dimensions needed to put the holes in the correct spots. My portion of the assignment took me about 3 hours to complete.

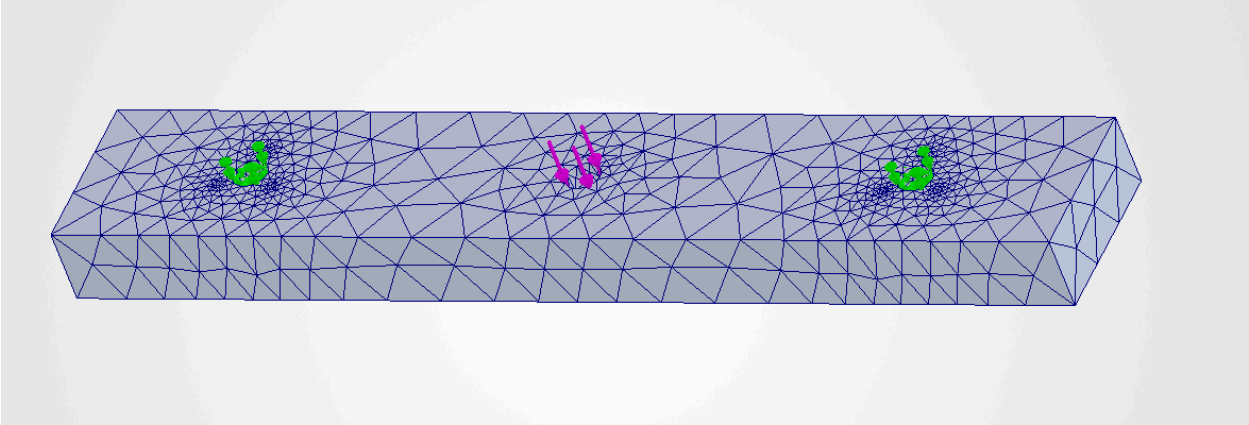
## FEA (Crosshead) - Levi Elliott

We are tasked with creating a FEA of the crosshead. The first thing I did was assign the external force of 2kn and the fixed positions.

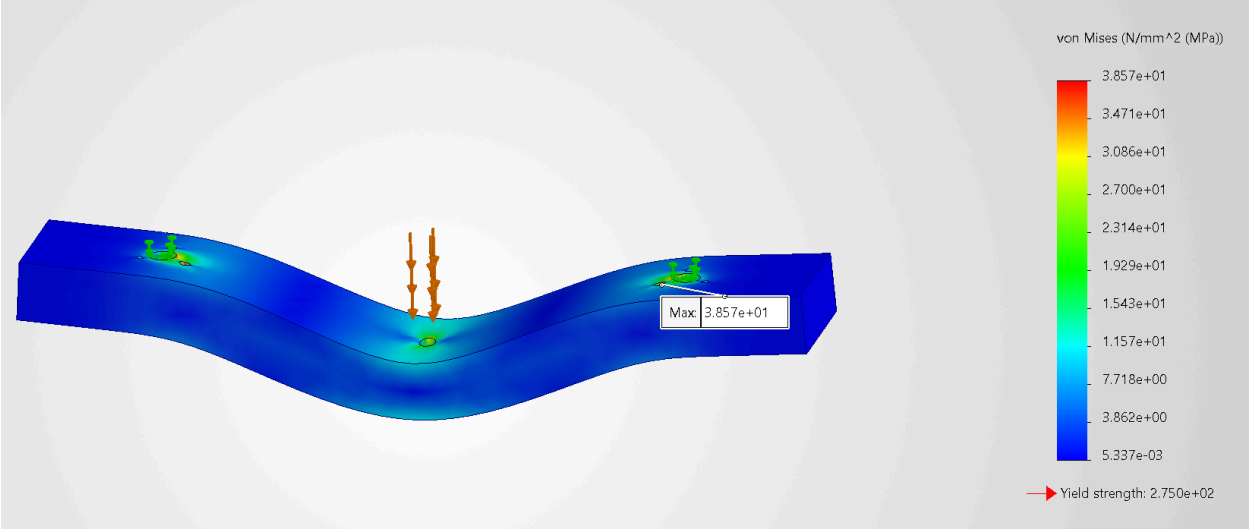


Then I created the mesh of the crosshead with the constraints.

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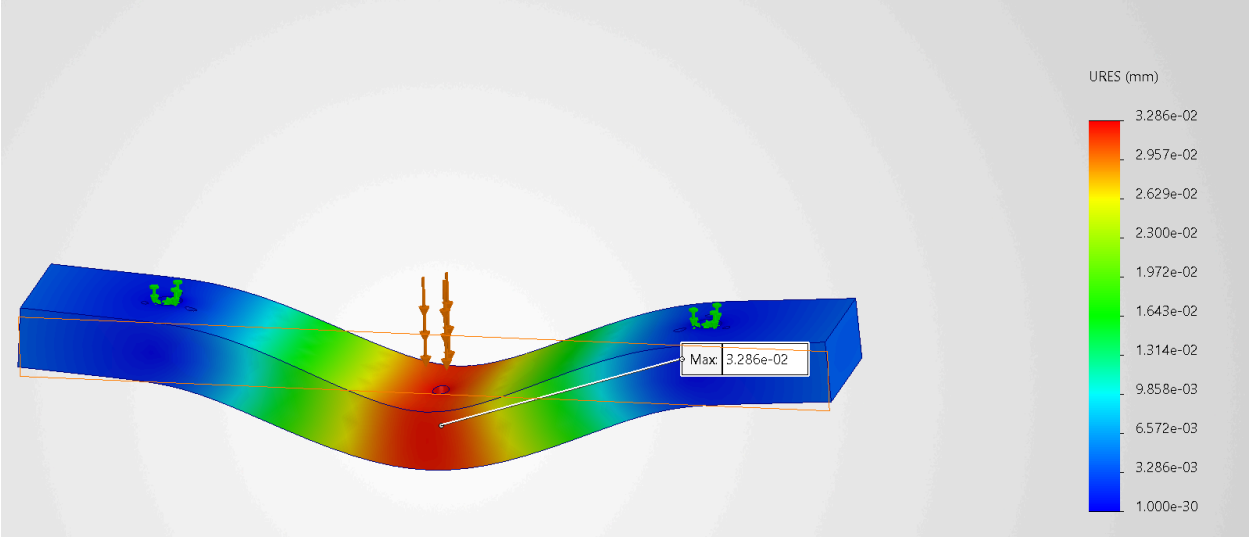


Then I ran the study, the max stress is 38.57 MPa. The max stress is located beside the lead screw holes.




The max vertical deflection is 0.03286 mm. This is located in the center of the crosshead. This is because the load of 2 kn is located at the center.

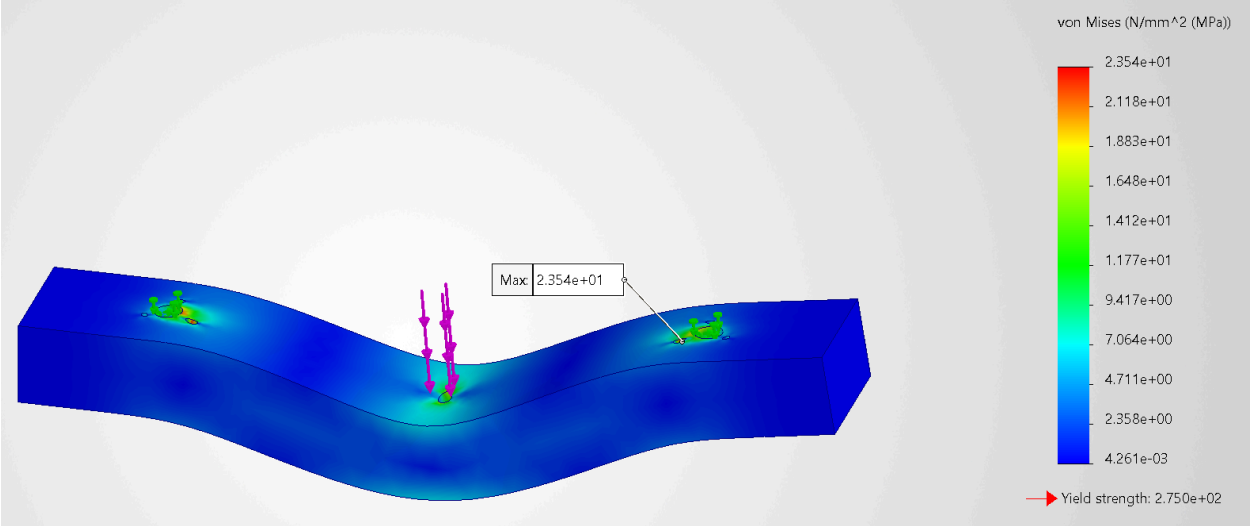
# Assignment 10: 2157 Students



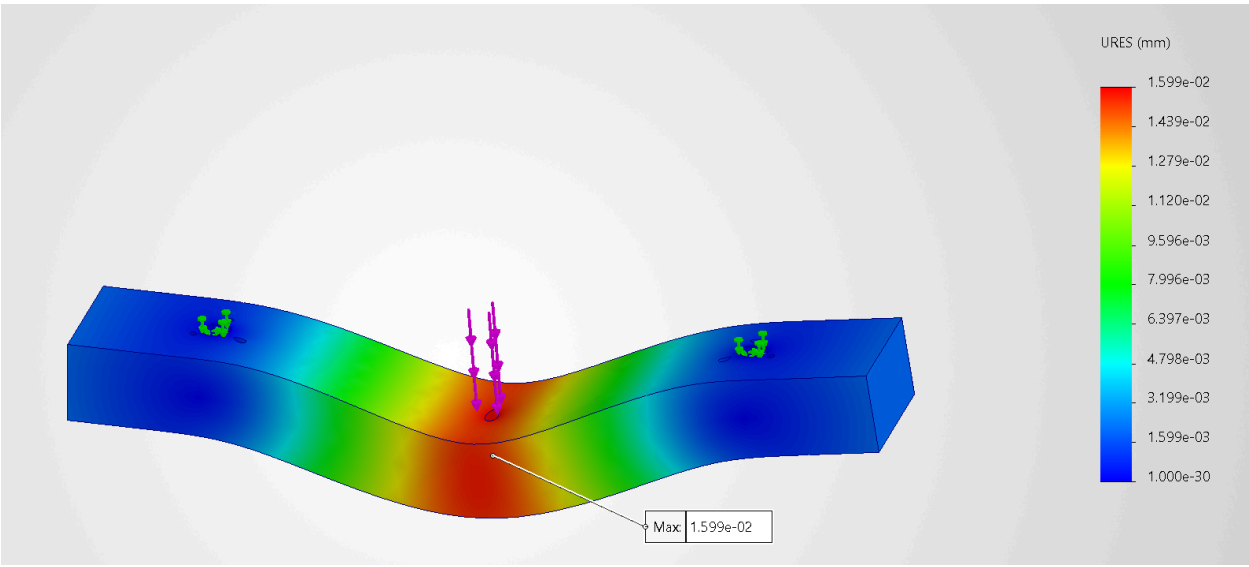

The load path of the crosshead is in the center where the force is applied. This means that the center of the crosshead takes on the most deflection vertically. The worst warping occurs at the center out to the lead screw holes. Our crosshead safety factor is 4, it did meet the safety factor. So this means I need to make some changes. For the second FEA I need to conduct I am going to increase the height of our crosshead from 22.63 mm to 30 mm.

Once I made the change to a height of 30 mm. The max stress is 23.54 MPa, which is located at the same spot as before beside the lead screws.

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The max vertical deflection is 0.01599 mm. This occurs at the same spot as the original, at the center of the part. The mass changed from  898.69 grams to 1191.37 grams after the height change.

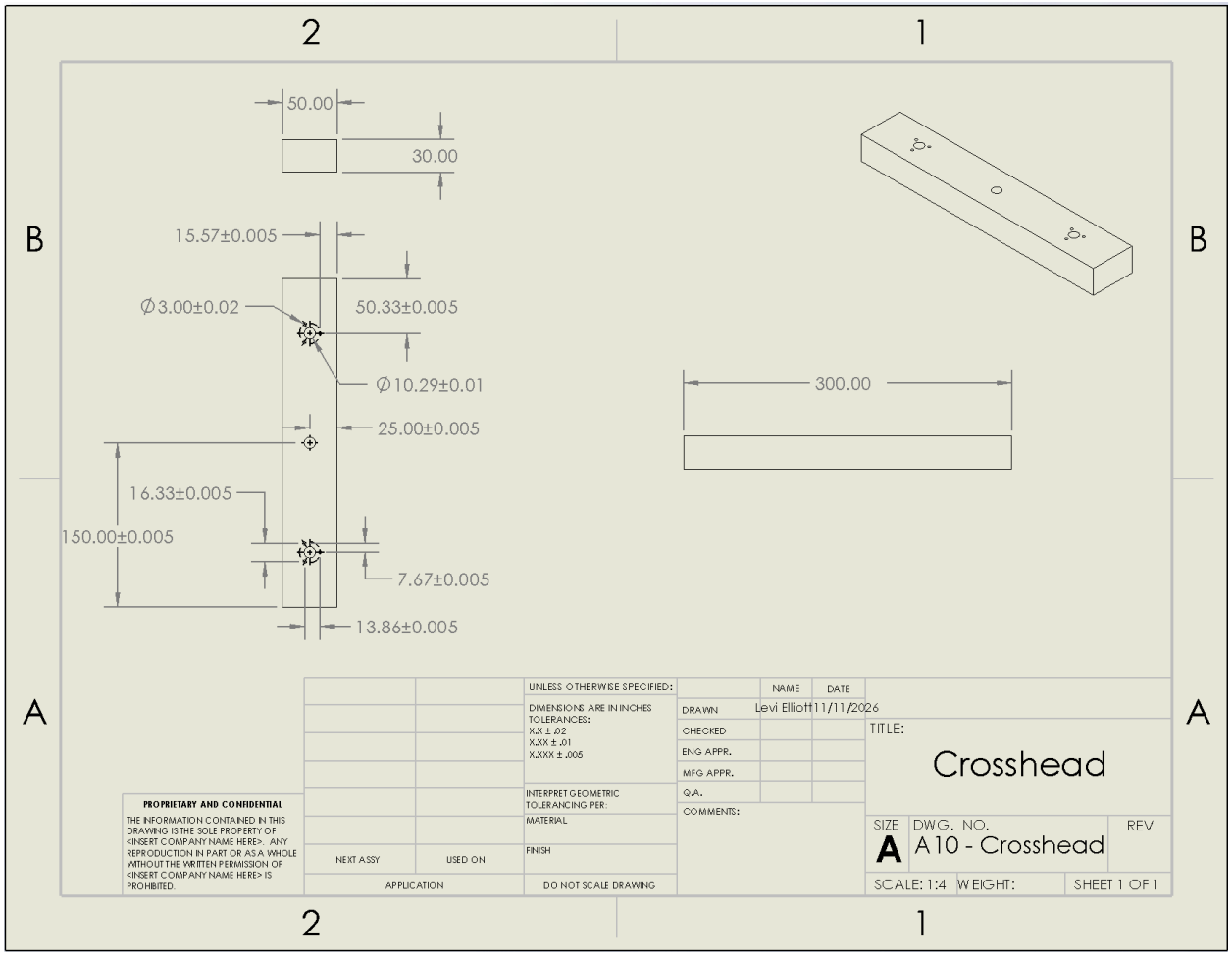



After I made the height change the max stress and deflection both decreased. The locations of the critical conditions did not change because it was just a height change. The crosshead now has a minimum safety factor of 12, so the part meets the constraints.

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## Drawing (Crosshead) - Levi Elliott

Then we are tasked with creating a detailed drawing of the crosshead.




A10 - Crosshead.SLDDRW